BIOCHEMISTRY CUMULATIVE EXAM

March 28, 1992

(20) 1. Aldolase catalyzes the following reaction along the glycolytic pathway:

Fructose-1,6-biphosphate (F) <--> Glyceraldehyde- 3-phosphate (G) + dihydroxy acetone phosphate (D).

$$F \leftrightarrow G + D; \quad K_{eq1} = 1.5 \times 10^{-3} \qquad (1)$$

Calculate the free energy ($\Delta G'$) under physiological conditions for the conversion of 10 mMoles of F to D and G if the starting concentration of F is 5 mM and the concentrations of products is 1 mM.

(40) 2. Triose phosphate isomerase catalyzes the following reaction:

$$G \leftrightarrow D; \quad K_{eq2} = 19 \qquad (2)$$

Show how you would calculate the free energy for the conversion of F to G + D in the presence of both enzymes.

(40) 3. Will the presence of triose phosphate isomerase shift equilibrium 1 to the right or the left if these reactions are near equlibrium along the glycolytic pathway? Explain how you arrived at your answer.

$$\Delta G = \Delta G^{o} + RT \ln \text{products/reactants}$$